# FIFA 2026 WORLDCUP

## MATCH OUTCOME PREDICTION PROJECT

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# PROBLEM STATEMENT

Football match outcomes are influenced by several complex factors such as:

- Team strength and FIFA rankings
- Player skill levels
- Historical match performance
- Environmental conditions

The goal of this project is to develop a predictive framework capable of estimating match outcomes for FIFA World Cup 2026 using historical football data and machine learning techniques.

This system integrates multiple datasets to analyze how team strength, player quality, and environmental conditions influence match results.

| Confederation | Number of slots | Number of slots |
|---|---|---|
| AFC | 8 | 1 |
| CAF | 9 | 1 |
| Concacaf | 6 (including Canada, Mexico and the USA as co-hosts) | 2 |
| CONMEBOL | 6 | 1 |
| OFC | 1 | 1 |
| UEFA | 16 | - |
| Total number of qualification slots for the FIFA World Cup 26 | 46 | 2 |

| Group A | Group B | Group C | Group D | Group E | Group F |
|---|---|---|---|---|---|
| A1 (Mexico) | B1 (Canada) | C1 | D1 (USA) | E1 | F1 |
| A2 | B2 | C2 | D2 | E2 | F2 |
| A3 | B3 | C3 | D3 | E3 | F3 |
| A4 | B4 | C4 | D4 | E4 | F4 |

| Group G | Group H | Group I | Group J | Group K | Group L |
|---|---|---|---|---|---|
| G1 | H1 | I1 | J1 | K1 | L1 |
| G2 | H2 | I2 | J2 | K2 | L2 |
| G3 | H3 | I3 | J3 | K3 | L3 |
| G4 | H4 | I4 | J4 | K4 | L4 |

# SCHEDULE OF FIFA2026!

11TH JUNE - 19TH JULY

48 TEAMS 12 GROUPS - TOTAL
2 TEAMS FROM EACH GROUP IS QUALIFIED - 24 TEAMS
TOP 8 THIRDPLACE TEAMS GETS QUALIFIED - 8 TEAMS
ROUND OF 32 - KNOCKOUTS
ROUND OF 16 - KNOCKOUTS
QUARTER FINALS - KNOCKOUTS
SEMIFINALS - KNOCKOUTS
FINALS - KNOCKOUTS

# DATA & DATA MANAGEMENT

International Match Results

- 49,000+ matches
- Match date
- Home and away teams
- Scores
- Tournament information

FIFA Rankings

- Team rankings
- Ranking points

Weather Data

- Temperature
- Humidity
- Wind speed

Data Management

- Data stored in SQLite database
- Managed using Python Pandas
- Enables efficient querying and integration.

Player Attributes

- Player ratings
- Technical abilities

# DATA PREPROCESSING

Data preprocessing was performed to ensure the dataset was clean, consistent, and suitable for analysis and machine learning modeling.

## Data Cleaning

Several steps were taken to improve data quality:

- Removed missing or incomplete values
- Checked and corrected inconsistent entries
- Ensured all numerical variables were properly formatted

These steps help prevent errors during analysis and model training.

## Data Standardization

To ensure consistency across datasets:

- Team names were converted to lowercase
- Extra spaces and formatting differences were removed
- Data types were standardized for numerical variables

This ensured that teams from different datasets could be correctly matched and merged.

## Feature Engineering

New variables were created to improve prediction performance:

- Ranking difference between home and away teams
- Total goals scored in a match
- Match outcome variable
  - Home win = 1
  - Draw = 0
  - Away win = -1

These engineered features help the model better understand relative team strength and  match dynamics.

## Data Integration

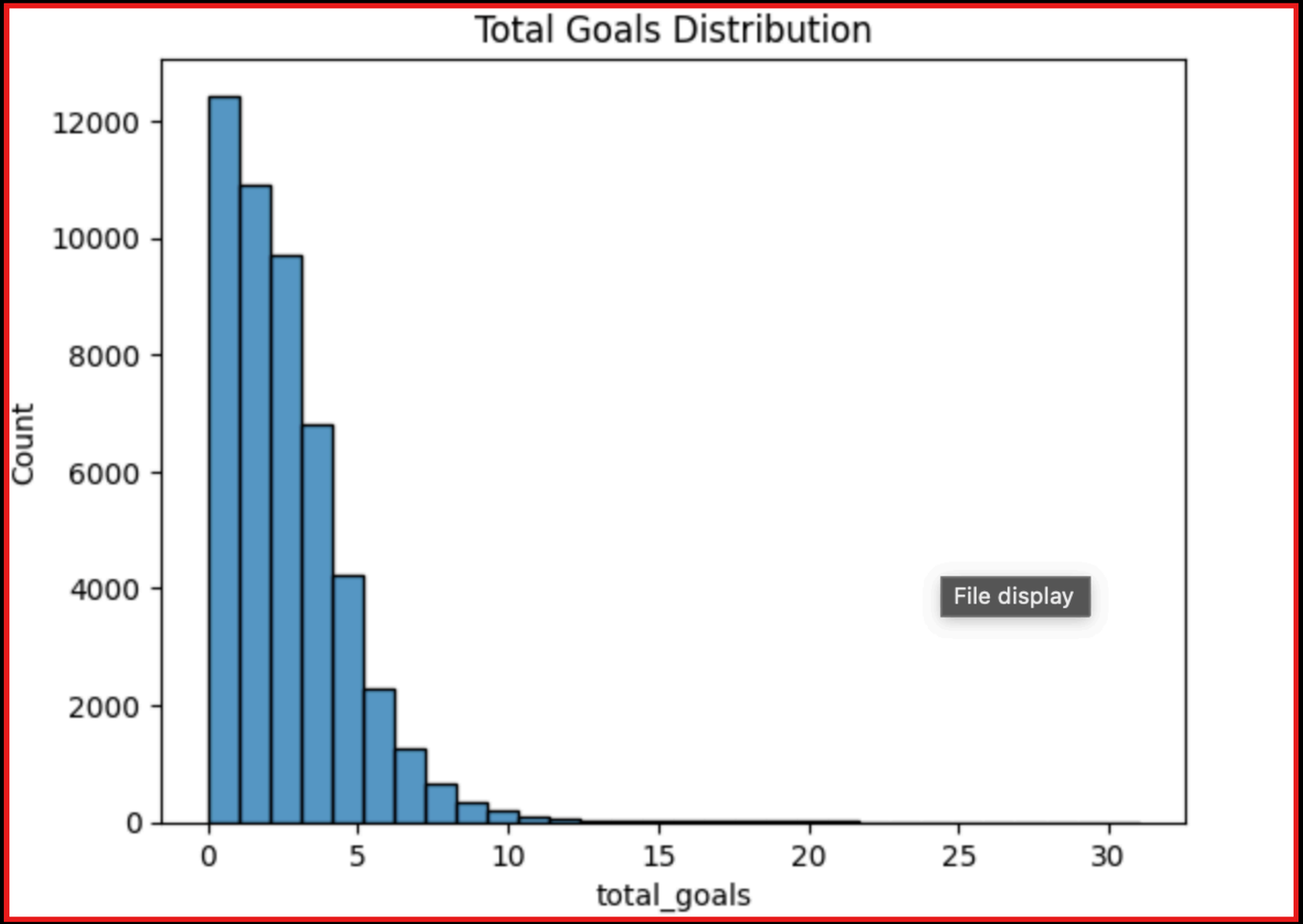
Multiple datasets were combined into a unified analytical dataset:

- Historical match results
- FIFA rankings
- Player attributes
- Weather data

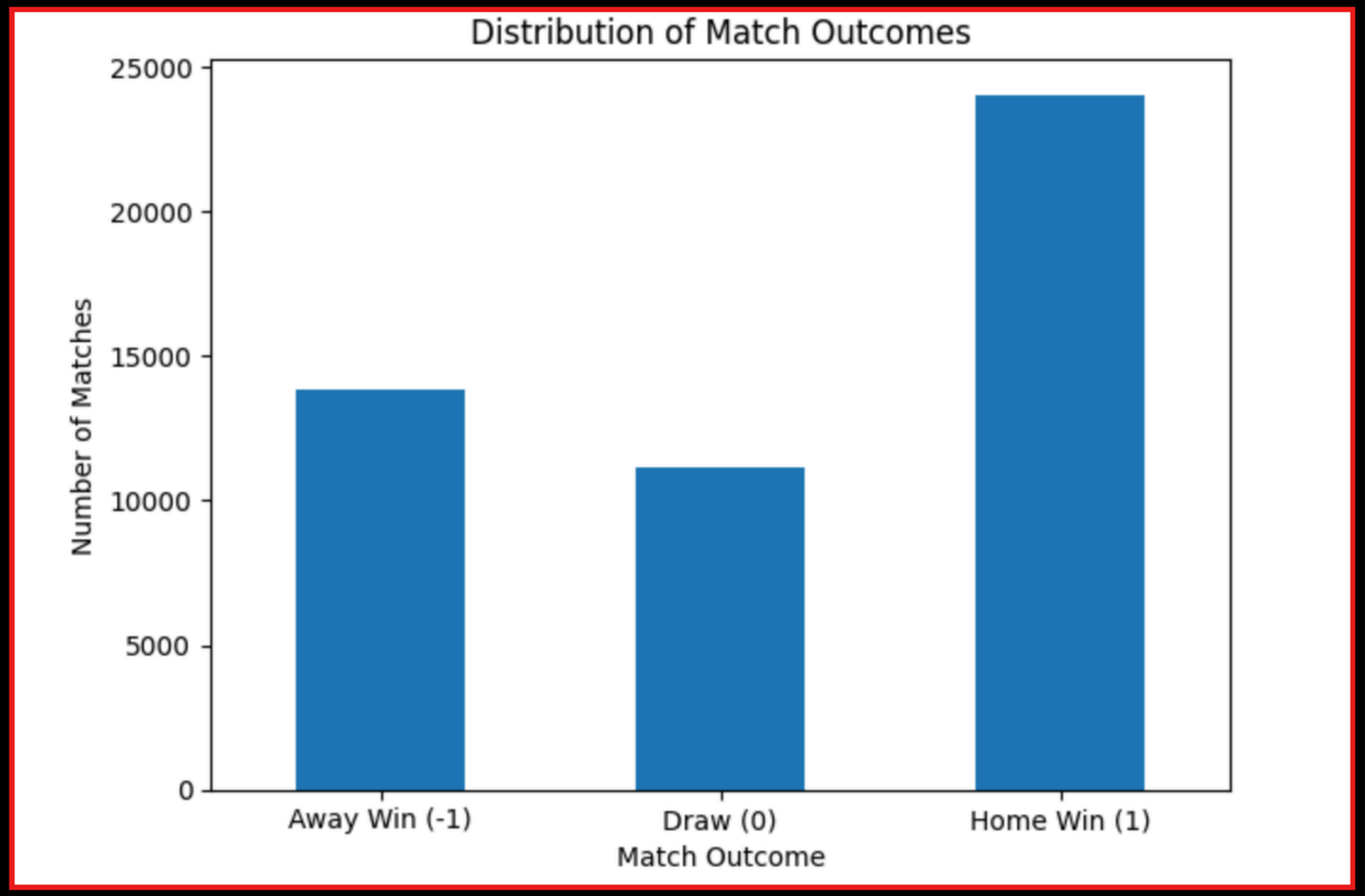
The integrated dataset provides a comprehensive view of factors influencing football match outcomes.

# EXPLORATORY DATA ANALYSIS (EDA)

- DATASET CONTAINS 49,000+ INTERNATIONAL MATCHES
- AVERAGE GOALS SCORED PER MATCH IS APPROXIMATELY 3
- MATCHES BETWEEN SIMILARLY RANKED TEAMS ARE MORE COMPETITIVE
- LARGE RANKING DIFFERENCES LEAD TO MORE PREDICTABLE OUTCOMES
- WEATHER VARIABLES SUCH AS TEMPERATURE AND HUMIDITY WERE ALSO ANALYZED TO EVALUATE ENVIRONMENTAL IMPACTS




THE DISTRIBUTION OF TOTAL GOALS IS HIGHLY RIGHT-SKEWED, WITH MOST MATCHES PRODUCING FEWER THAN FIVE GOALS. THE LONG RIGHT TAIL SUGGESTS POSSIBLE OVERDISPERSION RELATIVE TO A STANDARD POISSON MODEL.




THIS DISTRIBUTION SHOWS THAT HOME WINS ARE THE MOST FREQUENT OUTCOME, FOLLOWED BY AWAY WINS, WHILE DRAWS OCCUR LEAST OFTEN. THIS SUGGESTS A CLEAR HOME-FIELD ADVANTAGE, WHERE TEAMS ARE SIGNIFICANTLY MORE LIKELY TO WIN WHEN PLAYING AT HOME. THE IMBALANCE ALSO INDICATES THAT MATCH OUTCOMES ARE NOT EVENLY DISTRIBUTED ACROSS THE THREE CATEGORIES.

# MODELING & ANALYSIS

```python
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import roc_auc_score, roc_curve, auc, classification_report, confusion_matrix, accuracy_score
from sklearn.preprocessing import label_binarize

import statsmodels.api as sm
import statsmodels.formula.api as smf
from statsmodels.stats.outliers_influence import variance_inflation_factor
```

# MODELING FRAMEWORK

THE INTEGRATED DATASET ENABLES THE APPLICATION OF MACHINE LEARNING MODELS TO PREDICT MATCH OUTCOMES.

MODELS THAT CAN BE APPLIED INCLUDE:

- LOGISTIC REGRESSION
- RANDOM FOREST
- GRADIENT BOOSTING

TO ENSURE RELIABLE RESULTS, K-FOLD CROSS-VALIDATION CAN BE USED DURING MODEL TRAINING.

# MODEL EVALUATION METRICS

- ACCURACY
- CONFUSION MATRIX
- CLASSIFICATION REPORT
- ROC-AUC

FEATURE IMPORTANCE ANALYSIS CAN ALSO IDENTIFY KEY FACTORS INFLUENCING MATCH OUTCOMES.

# CHALLENGES & FUTURE WORK

## Challenges

- Incomplete player data for some teams
- Weather data availability limitations
- Team roster changes affecting predictions

## Future Work

- Incorporate live FIFA rankings
- Use player injury and squad data
- Implement advanced deep learning models

# CONCLUSION

In [315…

```
semifinals = make_next_round(quarterfinals_pred["winner"].tolist())

semifinals_pred = predict_knockout_round(
    semifinals,
    xgb_model=xgb_model,
    X_train=X_train,
    elo_strength=elo_strength,
    match_month=7
)

print(semifinals_pred[["home_team", "away_team", "winner"]])
```

```
   home_team away_team     winner
0     Brazil     Spain      Spain
1  Argentina   Uruguay  Argentina
```

In [316…

```
final = make_next_round(semifinals_pred["winner"].tolist())

final_pred = predict_knockout_round(
    final,
    xgb_model=xgb_model,
    X_train=X_train,
    elo_strength=elo_strength,
    match_month=7
)

print(final_pred[["home_team", "away_team", "winner"]])
print("Predicted Champion:", final_pred["winner"].iloc[0])
```

```
  home_team  away_team winner
0     Spain  Argentina  Spain
Predicted Champion: Spain
```

This project developed a data-driven framework to predict international football match outcomes by integrating multiple datasets such as historical match results, FIFA rankings, player attributes, and environmental data.

Through data preprocessing, exploratory data analysis, and machine learning modeling, the analysis identified important factors that influence match results. Key predictors included team strength indicators such as FIFA rankings and ranking differences, along with match context variables.

The exploratory analysis also revealed common patterns in football matches, including the low-scoring nature of games and the strong advantage of playing at home.
Using the trained prediction framework and historical performance data, the model was applied to simulate match outcomes for the FIFA World Cup 2026 tournament.

## Final Prediction

Based on the analysis and simulation results, the model predicts that Spain has the highest probability of winning the FIFA World Cup 2026.
This prediction reflects Spain's strong historical performance, high team ranking, and consistent competitive strength compared to other teams.

[316…

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    final,
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    X_train=X_train,
    elo_strength=elo_strength,
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)

print(final_pred[["home_team", "away_team", "winner"]])
print("Predicted Champion:", final_pred["winner"].iloc[0])
```

```
  home_team  away_team winner
0     Spain  Argentina  Spain
Predicted Champion: Spain
```

# SIMULATION

To predict the outcome of the FIFA World Cup 2026, a simulation-based approach was implemented using historical international football match data and team strength indicators. First, team strength was estimated using factors such as FIFA rankings, historical match performance, and ranking differences between competing teams. Using these features, a predictive model was trained to estimate the probability of one team winning against another. The simulation then recreated tournament matches by using these probabilities to determine the winner of each game. By running the tournament simulation multiple times, the model was able to observe which teams advanced through each stage most frequently. The team that won the highest number of simulated tournaments was identified as the most likely champion. Based on the simulation results, Spain emerged as the team with the highest probability of winning the FIFA World Cup 2026.

https://predicting-fifa-world-cup-2026.vercel.app